

LEARNING OBJECTIVES.

- *Calculate point estimates and credible intervals from a posterior distribution and give interpretations for the latter.*
- *Conduct hypothesis tests based on a posterior distribution.*
- *Calculate a prediction based on a posterior distribution.*

In the NCAA Division I men's basketball tournament, there are four regions with teams seeded 1 (the best) through 16 (the worst). In the first round, the 1 seed plays the 16 seed, the 2 seed plays the 15 seed, and so on. An upset (at least, by seed) occurs when a team with a better seed loses to a team with a worse seed. One upset that seems to occur with surprisingly high frequency is the 12 seed beating the 5 seed. In the 2025 version of the tournament, it happened twice (Colorado State beat Memphis and McNeese beat Clemson). Consider the parameter π , which represents the true probability that a 12 seed would beat a 5 seed in a hypothetical first round tournament game (or the proportion of times those upsets happen in the entire hypothetical population). Our task is to perform inference about π . Specifically, suppose we are interested in an interval estimate of π as well as testing the claim that more than one upset will happen on average each year.

Example 1. We will begin by performing frequentist inference. Give a point estimate and 95% confidence interval (with interpretation) for π . Then conduct a hypothesis test for the above claim. Include an interpretation of the p-value. For the data, use the fact that in the 40 years since the tournament went to this format there have been 57 of this type of upset.

Now consider the Bayesian approach, which starts by finding a posterior distribution.

Example 2. Suppose before we collected any data, we thought that π was definitely less than half, and that the most plausible value was somewhere between 0.25 and 0.3. Give a reasonable prior and then calculate the posterior using a Beta-Binomial model.

Example 3. Give a point estimate for π based on the posterior above.

Posterior Credible Intervals

Example 4. Calculate and interpret a 95% posterior credible interval for π .

Bayesian Hypothesis Testing: Idea

Example 5. Calculate the probability of the alternative hypothesis in our basketball example. Is that enough to convince you that the alternative is true?

Example 6. Suppose we had a two-sided alternative hypothesis instead. Could you still perform inference the same way?

Two-Sided Tests

Example 7. Using our posterior for π , discuss the evidence for the following hypotheses:

$$H_0 : \pi \in (0.2, 0.3), \quad H_a : \pi \notin (0.2, 0.3)$$

Bayes Factor

The Bayes Factor is a measure of how much our understanding of the parameter has evolved from the prior to the posterior, and is another piece of information that can be used to help make a decision about a hypothesis. Specifically, the Bayes Factor (BF) is the odds ratio for H_a :

As a ratio, it is meaningful to compare BF to 1. To this end, consider the following scenarios:

Example 8. Calculate the Bayes Factor for our original hypothesis test and interpret the results.

Posterior Prediction: Beyond posterior estimation and hypothesis testing, a third common task in a posterior analysis is to predict the outcome of new data. For example, based on our basketball posterior, how might we predict the number of 12-5 upsets we will see next year? What types of variability do we need to take into account?

Posterior Predictive Model

For the Beta-Binomial model, the pmf of Y' is

$$f(y'|Y = y) = \binom{n'}{y'} \frac{\Gamma(\alpha + y + \beta + n)}{\Gamma(\alpha + y)\Gamma(\beta + n)} \frac{\Gamma(\alpha + y + y')\Gamma(\beta + n + n' - y')}{\Gamma(\alpha + y + \beta + n + n')}$$

Posterior Analysis from MCMC: We can also use results from an MCMC simulation to do posterior estimation, hypothesis testing, and prediction.

Example 9. Use the R code provided to approximate the posterior distribution for our example and confirm the inferential results from the exact posterior.